ANIMATRONICS

OVERVIEW

Animatronics refers to a robotic device that emulates a human or an animal, or brings an inanimate object "to life." Applying leadership and 21st century skills, teams produce an animatronics device complete with an appropriate display. The animatronics device must use control technology in its performance and fulfill the requirements of the theme to communicate, entertain, inform, demonstrate and/or illustrate a topic, idea, subject, or concept. Sound, lights, and surrounding environment are to accompany the device. The entry must reflect the annual theme. The annual design problem is posted on the TSA website under *Themes & Problems*.

ELIGIBILITY

Three (3) teams of two to three (2-3) team members per state may participate.

TIME LIMITS

A. Up to five (5) minutes to set up. Four (4) teams to set up at once.

B. Up to five (5) minutes for the presentation.

   1. Presentation time begins when students present background information about the project and must conclude on or before the five (5)-minute time limit.
   2. Timekeeper will stop presentation at five (5) minutes.

C. Up to two (2) minutes to respond to questions from the judges.

ATTIRE

TSA competition attire is required for this event.

PROCEDURE

PRE-CONFERENCE

A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules.

B. Participants review the Use of Generative Artificial Intelligence (GenAI) found in the General Rules.

C. Participants review the design problem on the TSA website under *Themes & Problems*.

D. Participants concentrate their efforts on designing an animatronics device that uses control technology.

PRELIMINARY ROUND

A. Participants report to the time and place stated in the conference program to:

   1. Check in
   2. Sign up for a presentation/interview time

   No animatronics devices are submitted during this time.

B. Participants report for the presentation/interview at the selected demonstration time with the animatronics device and the documentation portfolio. Two (2) or three (3) team members are allowed to set up equipment, present the project, and participate in the event-specific interview.

C. Team must reclaim the documentation portfolio following the presentation and judging in order to have it for the semifinal round.

D. Judges independently assess the entries.

E. A list of twelve (12) semifinalists (in random order) is posted.

SEMIFINAL ROUND

A. Semifinalists report to the event area at the time and place stated in the conference program to receive an assigned presentation/interview time.

B. Semifinalist presentation/interviews follow the same procedure as in the preliminary round.

C. The top ten (10) finalists will be announced during the conference awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRELIMINARY ROUND

A. Documentation Portfolio:
   1. Documentation materials (comprising "a portfolio") are required and must be secured in a clear front report cover with the following single-sided, 8½" x 11" pages, in this order:
      a. Title page with the event title, the team identification number, the conference city and state, and the year; one (1) page
      b. Table of contents; pages as needed
      c. Student Copyright Checklist (see 2027 & 2028 Forms Appendix); one (1) page
      d. Resources and AI Reflection (see 2027 & 2028 Forms Appendix); maximum of four (4) pages

B. Starting position (resting) dimensions of the entry may not exceed 15" deep x 3' wide x 4' high. Device may extend beyond the dimensions of the display during the demonstration.

C. Exterior shell or skin is required. It must be removable in order to show the judges the internal components of the project.

D. Animatronic device must have at least three (3) separate movements that must include:
   1. Fluid power to aid in the movement of the animatronics device.
   2. Sound, lights, and sensors in the project model.
   3. Gearing systems, linkages, and/or cabling systems, etc., to aid in the movement of the device.

E. Control technology must be used during the performance.

F. A wet cell battery may not be used in the animatronics device.

G. Animatronics device may use AC power, but the team will only have access to an AC outlet during the presentation/interview.

H. Should the device suggest anything that is inappropriate by language, sound, or movement, immediate disqualification will result.

I. A team that fails to appear for its presentation forfeits evaluation.

SEMIFINAL ROUND

A. Semifinalists report at the time and place stated in the conference program to sign up for a presentation/interview time.

B. Semifinalists report at the assigned time and place for the presentation/interview.

C. All regulations from the preliminary round apply to the semifinal round; including bringing a copy of the documentation portfolio.

D. Judges evaluate the entries.

E. The top ten (10) finalists are announced at the awards ceremony.

EVALUATION

A. Device

B. Presentation/interview

C. Documentation portfolio

Refer to the official rating form for more information.

GENERATIVE ARTIFICIAL INTELLIGENCE

Generative Artificial Intelligence (GenAI) is permitted for this event. Refer to the Use of Generative Artificial Intelligence (GenAI) for Competitive Events found in the General Rules and Regulations.

NOTES

Learn more about animatronics by visiting the following:

www.roborobotics.com/Animatronics/Animatronics.html
www.animatronicdatabase.com
www.techanimatorsalliance.com

STEM INTEGRATION

This event aligns with the STEM (Science, Technology, Engineering, and Mathematics) educational standards.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Amusement park robotics maintenance engineer
- Electronics technician
- Film industry special effects engineer
- Industrial designer
- Toy developer

Participant/Team ID# _____

ANIMATRONICS

2027 & 2028 OFFICIAL RATING FORM

HIGH SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

☐ Skins/shells are removable
☐ Interior skeleton and mechanism are accessible for inspection
☐ Battery meets regulations
☐ Entry meets initial "resting" position dimensions
☐ Documentation portfolio submitted
☐ ENTRY NOT EVALUATED

MODEL APPEARANCE (60 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Utilization of Annual Theme (X3)	Model doesn't reflect the posted annual theme.	Model makes some attempt at meeting the posted theme but leaves some doubt.	Model clearly depicts the annual theme.	
Creativity, Aesthetics and Artisanship, Originality (X3)	Model lacks creativity; very few or no design principles are integrated in the model; work is unorganized and/or sloppy; model seems to be an afterthought and/or thrown together; model lacks imagination, originality, and artistic detail.	Some elements of creativity are evident, and most essential design principles are included and used somewhat effectively; some layout and design principles are integrated into the model, and aesthetics are adequate; model is somewhat innovative.	Model exudes creativity; essential design principles and elements are integrated; there is exemplary use of layout and design principles; artistic and aesthetic values are incorporated; model is inspiring, inventive, and resourceful.	
			MODEL APPEARANCE SUBTOTAL (60 points)	

MODEL FUNCTION (70 points)				Record scores in the column spaces below.
Skin and skeletal function: There is no point value for the skin and skeletal function of the animatronics model. Model's skin must be removable in order to reveal skeletal function and mechanics located beneath the skin. If the skin is not removable then the entry will not be evaluated.				
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Sound Inclusion (X1)	There is little or no sound included, or the design suggests that the inclusion of sound was an afterthought to the model.	Sound is included, and it somewhat contributes to the overall function of the model.	Inclusion of sound is creative and effectively contributes to the design and performance of the model.	

MODEL FUNCTION (70 points) – continued				
Light Inclusion (X1)	There is no light included, light is minimal, or the design suggests that the inclusion of lights was an afterthought to the model.	Light is included, and it somewhat contributes to the overall function of the model.	Inclusion of a light creatively and effectively contributes to the model's design and performance.	
Sensor Inclusion (X2)	No sensors are included, sensors are included minimally, or the design suggests that the inclusion of sensors was an afterthought to the model.	Sensors are included, and they somewhat contribute to the overall function of the model.	Inclusion of sensors (and the interactivity that sensors allow) in the model is creative and effectively contributes to its design and performance.	
Control Technology (X1)	No control technology or little control technology is used during the performance.	Some basic control technology is used during the performance.	Advanced control technology is used during the performance; model is fully autonomous.	
Fluid Power System Inclusion (X1)	No fluid power system is included or a fluid power system is included, but it functions inadequately or not at all.	A fluid power system is included, and it contributes somewhat to the overall function of the model.	Inclusion of a fluid power system(s) and the fluidity of movement that this system(s) provides in an animatronics model creatively and effectively contribute to the model's design and performance.	
Use of Gears, Linkages, Cabling, etc. (X1)	No use of gears, linkages, cables or the use of gears, linkages, cabling, etc. is minimally apparent or improperly incorporated into the model; team shows little understanding of how to properly use these systems in the model.	Most gears, linkages, cabling systems, etc. are incorporated and used properly in the model; there is evidence of an adequate understanding of the systems.	Efficient and varied use of gears, linkages, cabling systems, etc. is apparent and properly incorporated in the model; there is evidence of a complete understanding of these systems.	
			MODEL FUNCTION SUBTOTAL (70 points)	

PRESENTATION/INTERVIEW (30 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Organization and Knowledge (X1)	Team seems unprepared and unorganized for the presentation/ interview, with an illogical explanation of the project; team members seem to have little understanding of the concepts in their project; vague interview answers are provided.	Team is prepared for the interview and is somewhat organized in its presentation to judges; team's presentation is somewhat logical and/or clear; team members have a general understanding of the concepts discussed and answer questions adequately.	Team's presentation/interview with judges is well organized; interview is concise and logical, with a clear explanation of the development of the project; evidence is clear that team members have a thorough understanding of the concepts discussed; they answer questions thoroughly.	
Articulation (X1)	Communication of the design process is unclear, unorganized, and or illogical; leadership and/or 21st century skills are not evident.	Communication of the design process is somewhat logical and clear; leadership and/or 21st century skills are somewhat evident.	Communication of the design process is clear, concise, and logical; leadership and/or 21st century skills are clearly evident.	
Team Participation (X1)	Only one person in the group communicates with judges; there is little or no participation from other team members.	Team members all participate to some extent and seem to understand the concepts.	Team members seem to fully understand the concepts and share an equal role in the interview.	
			PRESENTATION/INTERVIEW SUBTOTAL (30 points)	

DOCUMENTATION PORTFOLIO (20 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Portfolio Components (X1)	Portfolio is unorganized and/or is missing three (3) or more components.	Portfolio has most components and is generally organized per the order of the event guide listing; it has sufficient content.	All components are included in the portfolio; content and organization per the order of the event guide listing are excellent.	
Resources and AI Reflection (X1)	Form is incomplete or vague and does not clearly explain whether or how AI was used, showing little connection to the submission or understanding of effectiveness and ethics.	Form is complete and clearly explains AI use or the lack thereof with connections to the submission or theme/topic (when applicable), addressing effectiveness and ethical considerations with adequate analysis.	Form is complete, concise, and specific, critically evaluating how AI or non-AI tools shaped decisions and outcomes, with strong alignment to the submission or theme/topic (when applicable) and ethical reasoning.	
			DOCUMENTATION PORTFOLIO SUBTOTAL (20 points)	

Rules violations (a deduction of 20% of the total possible points in the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____	

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary. **TOTAL (180 points)**	

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

ANIMATRONICS
EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

A. Event coordinator
B. Assistant for check-in, one (1)
C. Timekeeper, one (1)
D. Doorkeeper, one (1) for each room of the preliminary heats
E. Judges, two (2) or more

MATERIALS

A. Coordinator's packet, containing:
    1. Event guidelines, one (1) copy for the coordinator and for each judge
    2. TSA Event Coordinator Report
    3. Stick-on labels for entries, as needed
B. Measuring tape
C. Tables for presentation
D. Table and chairs for judges
E. Access to AC power for presentation/interview only

RESPONSIBILITIES

AT THE CONFERENCE

A. Attend the mandatory coordinator's meeting at the designated time and location.
B. Report to the CRC room and check the contents of the coordinator's packet.
C. Review the event guidelines and check to see that enough judges/assistants have been scheduled.
D. Inspect the area or room in which the event is to be held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.
E. At least one (1) hour before the event is scheduled to begin, meet with judges and review the time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the CRC event manager before the event begins.
F. Logistics/Time management procedure: Four (4) tables for entry presentations. Have four (4) teams set up simultaneously. One (1) stays and three (3) leaves, waiting on their turn. After each team presents to the judges, each team takes their entries out, while the next teams enter and set up their entries. After all four (4) teams have presented, the next four (4) is invited in to set up their entries in the timed five (5) minutes. Repeat until done.
G. Time keeper starts the clock when the presentation begins and stops the presentation at five (5) minutes. No extension is allowed. Judges are allowed up to two (2) questions or two (2) minutes after the five (5)-minute presentation. Encourage the judges to score during and no more than two (2) minutes after any questions. Roll of the timekeeper is very critical in this event.
H. Ensure the judges have access to the online judging system.

PRELIMINARY ROUND

A. Check in participants at the time and place stated in the conference program. During check-in, participants only sign up for a presentation time and do not submit their entry.
B. If check-in is on the first day of the conference, requirements for attire do NOT apply during check-in.
C. As participants sign up for a presentation/interview time, notify them that they are to report fifteen (15) minutes prior to their scheduled time.
D. Late participants and/or entries are considered on a case-by-case basis and only when lateness is caused by events beyond the participant's control.
E. In order to compete, participants must be on the entry list or must have approval from the CRC.
F. Oversee the presentation/interviews.
G. Return the documentation portfolio to the team following the presentation/interview and judging.

H. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and CRC manager to determine either:

   1. To deduct twenty percent (20%) of the total possible points in this round or
   2. To disqualify the entry

   The event coordinator, judges, and CRC manager must initial either of these actions on the rating form

I. Following the preliminary heats, the judging system will tabulate the scores; twelve (12) semifinalists will be posted.

J. If necessary, manage security and the removal of materials from the area.

SEMIFINAL ROUND

A. Report at the time and place posted in the conference program to assign semifinalists a presentation/ interview time.

B. Using the same official rating form for the semifinalist, judges asses the semifinalist presentation and determine the ten (10) finalists.

C. Discuss rule violations (e.g. 20% deduction, disqualification) and have all relevant parties initial the rating form.

D. If necessary, manage security and the removal of materials from the area.